

A Survey on Lightweight Cryptographic Algorithms in IoT

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Abstract: *The Internet of Things (IoT) will soon penetrate every aspect of human life. Several threats and vulnerabilities are present due to the different devices and protocols used in an IoT system. Conventional cryptographic primitives or algorithms cannot run efficiently and are unsuitable for resource-constrained devices in IoT. Hence, a recently developed area of cryptography, known as lightweight cryptography, has been introduced, and over the years, numerous lightweight algorithms have been suggested. This paper gives a comprehensive overview of the lightweight cryptography field and considers various popular lightweight cryptographic algorithms proposed and evaluated over the past years for analysis. Different taxonomies of the algorithms and other associated concepts were also provided, which helps new researchers gain a quick overview of the field. Finally, a set of 11 selected ultra-lightweight algorithms are analyzed based on the software implementations, and their evaluation is carried out using different metrics.*

Keywords: *Lightweight cryptography, Lightweight cryptography algorithms, IoT security, Cryptography in IoT, Lightweight cryptography in IoT, Ultra-lightweight algorithms.*

1. Introduction

IoT supports ubiquitous computing, which contains devices embedded with microprocessors, microcontrollers, and other components to provide enhanced services. The devices part of the IoT system can be divided into two categories. One is the high-end devices, which contain devices like PCs, laptops, tablets, etc., and the second is the low-end devices, which contain sensors, actuators, RFID tags, etc [1]. These devices often need more resources like processing power, RAM, ROM, battery life, etc. So, there is a need to choose the communication protocols and other underlying technologies carefully so that the performance of these devices does not hinder the user experience [2]. The data communicated by these resource-constrained devices in IoT must be protected to prevent unauthorized access or damage to the IoT environment [3, 4]. So, it is inherent that various security mechanisms need to be considered to find the best possible solution to secure those devices [5].

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In the above table, NR represents several rounds in the cipher and KS represents the key size of the cipher in bits. The study of various attacks on the ciphers reveals that the current trend of attacking the ciphers uses differential fault analysis and hybrid attacks.

5. Conclusion

IoT amalgamates existing and new technologies with broad applications in many areas. The devices used for sensing and communication are often resource-constrained. Hence, there is a need to use existing LWC algorithms or develop new algorithms based on the application to provide security and privacy. This paper provides a brief survey of the existing LWC algorithms and a taxonomy that aids new researchers in gaining insight into them. This paper also categorizes the existing LWC algorithms into ultra-lightweight, low-cost, and lightweight algorithms, which is not seen in previous research. Finally, based on the analysis of LWC algorithm implementations, the best software-oriented algorithms in the ultra-lightweight category were identified as PRIDE, ITUbee, and IDEA. As these algorithms are still vulnerable to different attacks, the future research direction involves designing a new LWC algorithm that outperforms them and resists different attacks.

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